

MEMVERSE: Architecture, Pipeline & Operational Workflow

Detailed Technical Specification of the Zero-Trust Gateway, 12-Stage Pipeline, and Multimodal Data Flow

1. System Architecture Overview

The MEMVERSE platform is structured as a three-tier decoupled architecture:

- Client Layer (Frontend):** A single-page application built with React and Vite, hosting the conversational interface, Identity Vault, biometric scanner, and real-time security trace inspection drawer.
- Security Gateway Layer (Backend):** A high-throughput FastAPI service hosting the 12-stage security engine, policy evaluator, cryptographic key manager, and Merkle-style hash ledger.
- External Model Layer (Inference Engine):** Cloud-hosted NVIDIA NIM endpoints (e.g., Llama-3.3-70B-Instruct, Nemotron) that receive strictly sanitized payloads and return streaming responses.

2. The 12-Stage Zero-Trust Execution Pipeline

Each user interaction executes synchronously across twelve discrete processing stages:

Stage	Component	Operational Specification
01	Request Ingestion	Captures inbound prompt, session token, purpose declaration, and target model identifier.
02	Sensitive Entity Detection	Scans input for Named Entities (NER) and regex patterns across PII categories (names, emails, IDs, locations).
03	Poisoning & Injection Defense	Evaluates prompt against heuristic and semantic injection patterns to detect instruction override attempts.
04	Memory Retrieval	Queries local encrypted memory store for candidate records associated with the user profile.
05	Policy Matrix Evaluation	Applies policy matrix v1.4 to determine field-level actions (SUPPRESS, GENERALIZE, REDACT, or ALLOW).
06	Context Transformation	Replaces detected sensitive tokens with coarse-grained categorical bands and generalized identifiers.
07	Passport Scope Verification	Validates memory passport integrity hash, consent status, destination binding, and TTL expiration.
08	Approved Context Synthesis	Assembles the final sanitized context block and system prompt. Raw data is excluded.

Stage	Component	Operational Specification
09	Egress Validation Gate	Performs final automated boundary scan to guarantee zero prohibited fields exit the trusted domain.
10	External Model Execution	Dispatches sanitized prompt to NVIDIA NIM via secure SSE transport and receives streaming tokens.
11	Response Ingress Validation	Inspects incoming completion tokens to prevent memory leakage or malicious output reflection.
12	Audit Ledger Receipt	Appends a cryptographic SHA-256 receipt containing timestamp, event hash, and previous hash link.

3. Multimodal Ingestion Specifications

3.1 Structured Document Processing (PDF / DOCX)

When a document (e.g., an academic grade sheet or employment certificate) is uploaded:

- The server parses text and tabular data using PyPDF and PDFPlumber.
- Automated pattern recognizers detect registration numbers, marks, and signatures.
- Sensitive values are generalized into contextual performance tiers before ingestion into the user's active memory passport.

3.2 Biometric Face Image Ingestion

When photographic inputs are provided:

- Facial landmark detection executes locally in the client browser using `face-api.js`.
- The client presents an explicit consent confirmation dialog displaying the cropped region.
- EXIF data, GPS coordinates, and raw camera metadata are stripped before any downstream processing.

4. End-to-End User Operational Flow

Phase	User Action	System Execution & Output
1. Setup	Connects to platform or uploads identity document / image	Generates AES-256 encrypted memory records and issues active Memory Passport with TTL.
2. Interaction	Enters prompt in conversational interface	Gateway executes 12-stage pipeline, redacts PII, and constructs approved context block.
3. Response	Receives streaming model completion	NVIDIA NIM generates tailored output using sanitized background without access to raw PII.
4. Inspection	Opens Security Trace Drawer	Visualizes stage execution timings, transformed field mappings, and egress payload boundaries.
5. Audit	Views Event Ledger	Inspects tamper-evident SHA-256 receipts verifying provenance and non-repudiation.

5. Security Guarantees

1. **Client-Model Isolation:** The frontend client never possesses or transmits external model API keys.
2. **Fail-Closed Design:** Any failure in passport validation, consent check, or egress scanning immediately aborts the model call.
3. **Cryptographic Non-Repudiation:** Hash-linked ledger chains ensure that all memory access events are permanently auditable and tamper-evident.